

Flow Matching Algorithm for Generative Modelling: Basics

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Generating with Flow Models

- Firstly, a brief introduction of flows and velocity models (the basic tools used in this framework) is provided.
- Then, generation via flow models (continuous normalizing flows (CNF)) is introduced.
- After that, Flow Matching algorithm is presented as an optimization of CNFs.
- Finally, some considerations and possible improvement to the algorithm are presented,

Flows

- What is a flow? A Flow is a time dependent diffeomorphism defined as :

$$\psi : [0, 1] \times \mathbb{R}^d \rightarrow \mathbb{R}^d \quad (1)$$

- That means that for every continuous timestep $t \in [0, 1]$ we have a mapping :

$$\psi_t : \mathbb{R}^d \rightarrow \mathbb{R}^d \quad (2)$$

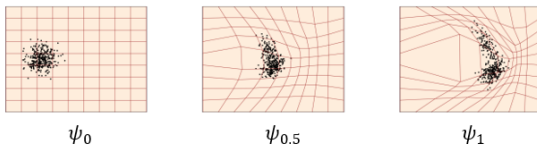


Figure: Visualization of ψ_t at different timesteps (brown square grid) applied to some points

Velocity Fields

- The flow ψ is characterized by a time dependent velocity field $u : [0, 1] \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ and the initial condition on ψ_0 :

$$\frac{d}{dt}\psi_t = u_t(\psi_t(x)) \quad (3)$$

$$\psi_0(x) = x \quad (4)$$

- Eq. (3) tells us that the Velocity model at time t expresses the instantaneous "change" of the mapping.
- Eq. (4) says that the first map of the flow (ψ_0) doesn't have any effect.

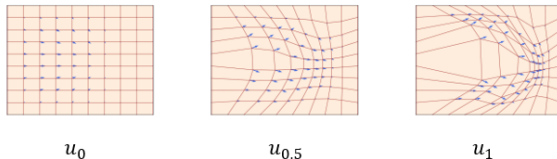


Figure: The blue arrows represent the velocity field at different time steps

Flow as generative models

- The goal of generative models is to learn a transformation that maps points $x_0 \sim p$, where p is a known and tractable distribution (standard Gaussian), to points $x_1 \sim q$, where q is the unknown data distribution underlying the observed samples.
- A flow model is a Markov Process defined by applying a flow ψ_t to the Random Variable X_0 .
- Eq. (4) tells us that, at time $t=0$, the model will not introduce any change:

$$\psi_0(X_0) = X_0 \sim p \quad (5)$$

- Finally, at $t = 1$, we would like the flow ψ_1 to produce an RV X_1 that distributes, at least approximately, as q :

$$\psi_1(X_0) = X_1 \sim q \quad (6)$$

- Applying ψ_t to X_0 we get X_t :

$$\psi_t(X_0) = X_t \quad (7)$$

- We denote as p_t the exact (unknown) distribution of the generic X_t :

$$X_t \sim p_t \quad (8)$$

- p_t , called Probability Path, is a time-dependent probability defined in $t \in [0, 1]$.
- p_t defines how the flow model "morphs" the initial RV X_0 over time.

Probability Paths (2)

- We can visualize the tie that exists between a velocity model and a probability path :

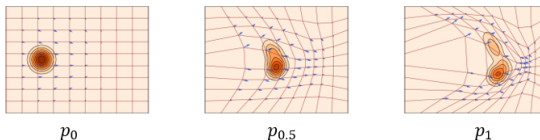


Figure: The Red-Yellow contour represents the PDF of the probability path at different time steps; note that the velocity model (blue arrows) morph the space (defining the flow model ψ) and, consequently, the origin probability defined at p_0

- Formally, we say that u_t Generates p_t if $X_t = \psi_t(X_0) \sim p_t$ for all $t \in [0, 1]$, ψ_t being the flow model described by u_t (see Eq. 3).

Velocity model estimation

- From Eq. (6) and Eq. (8) we deduce that :

$$\psi_1(X_0) = X_1 \sim p_1 \approx q \quad (9)$$

- Our goal is then to find a "good" velocity model u_t that defines a flow model ψ_t that produces a probability path p_t s.t. data is likely to come from distribution p_1 which is that is an approximation of q which defines the real generating mechanism of our dataset.
- We can assess the "goodness" of the velocity model by computing the log-likelihood of p_1 over data points x_1 ; by assuming p_1 is parametrized by a set of parameters θ that characterize the velocity model u_t^θ , we get the log-likelihood:

$$\ell(\theta; x) = \sum_{x_1 \in x} \log(p_1^\theta(x_1)) \quad (10)$$

Velocity model estimation (2)

- We compute the maximization that will result in $\hat{\theta}$:

$$\hat{\theta} = \max_{\theta} \ell(\theta; x) \quad (11)$$

- So that we get our trained velocity model $u_t^{\hat{\theta}}$ that we can use to generate random variables.
- In the following, an example of generation, by means of Euler's Method to solve the ODE defined in Eq. (3), of any RV X_t using the estimated velocity model:

$$X_{t+h} = X_t + hu_t^{\hat{\theta}}(X_t) \quad (12)$$

- where h is the difference between two chosen time steps.

Generation of a new data point

- By using Eq. (12) we can finally compute $\tilde{X}_1$, a numerical approximation of X_1 as (Note that $X_0 \sim p$):

$$\tilde{X}_1 = X_0 + u_0^{\hat{\theta}}(X_0) \approx \psi_1(X_0) = X_1 \quad (13)$$

- It follows, from Eq. (9) :

$$\tilde{X}_1 \sim p_t \approx q \quad (14)$$

- We finally managed to generate a sample from a approximate distribution of q ; the goodness of the approximation depends on the choice of the parameters $\hat{\theta}$ (Model training) and the numerical approximation of the ODE's solution; we chose Euler's Method for its simplicity, but there are more sophisticated methods)

Note on Euler's Method

- Euler's Method approximation is better if we compute $\tilde{X}_0$ by smaller steps h (Note that in Eq. (13) $h = 1$).
- The smaller the step h the better the approximation, but at an higher computational cost.
- A better generation of $\tilde{X}_1$ would be ($h = 0.1$ in this case) :

$$\begin{aligned}\tilde{X}_{0.1} &= X_0 + 0.1 u_0^{\hat{\theta}}(X_0), \\ \tilde{X}_{0.2} &= \tilde{X}_{0.1} + 0.1 u_{0.1}^{\hat{\theta}}(\tilde{X}_{0.1}), \\ &\dots \\ \tilde{X}_1 &= \tilde{X}_{0.9} + 0.1 u_{0.9}^{\hat{\theta}}(\tilde{X}_{0.9}).\end{aligned}\tag{15}$$

- We sample X_0 from a known distribution of our choice; usually that's a multivariate standard normal of dimension d .

$$X_0 \sim \mathcal{N}_d(0, I_d)\tag{16}$$

Drawbacks of Generation with Flow Models

- 1 This is a great method but the estimation of parameters $\hat{\theta}$ is not trivial, and often results in computational inefficiencies.
- 2 Methods to compute log-likelihoods are based on complicated solvings of several ODE equations.
- 3 Here's where Flow Matching Algorithm comes in our help, providing an easy and scalable way to train a good velocity model $u_t^{\hat{\theta}}$.

Flow Matching Algorithm

- Flow matching design is born from the question: "How can we compute the velocity field without having to compute the log-likelihood? What other optimization problem should we design to derive $u_t^{\hat{\theta}}$?"
- We can rethink the optimization problem defined by Eq. (11) as finding the best set of parameters θ that minimizes the Flow Matching Loss :

$$\mathcal{L}_{FM}(\theta) = \mathbb{E}_{t, p(x)} \left\| u_t^{\theta} - u_t \right\|^2 \quad (17)$$

$$\hat{\theta} = \min_{\theta} \mathcal{L}_{FM}(\theta) \quad (18)$$

- Where u_t^{θ} is a parametrized learnable velocity field and u_t is the unknown velocity field that generates some "good" target probability path p_t .

Flow Matching Algorithm (2)

- What's the difference with the optimization problem defined by Eq. (10)? The main one is that in Eq. (17),(18) we do not focus only on the last step of the target probability path p_1 , but we target the whole path p_t .
- Although this may sound unreasonable, we will discover how this to be much more efficient.

Conditional Flow Matching (CFM)

- Flow matching loss, as stated in Eq. (17), requires us to know the true velocity field u_t .
- This is intractable in practice: we don't know p_t or u_t , and we have no easy way to derive them.
- The authors of the FM algorithm proposed to work using conditional paths; we define the conditional probability path $p_t(x|x_1)$, where x_1 is a sample point from our dataset.
- We constrain $p_t(x|x_1)$ to meet the following conditions :

$$p_0(x|x_1) = \mathcal{N}(0, 1) \quad (19)$$

$$p_1(x|x_1) = \mathcal{N}(x_1, \sigma^2 I), \quad 0 < \sigma < \epsilon \quad (20)$$

- Eq. (19) states that, for any x_1 , p_0 should be the initial distribution p (Coherently with Eq. (5)) and Eq. (20) states that, for any x_1 , p_1 should be a Gaussian distributions with parameters $\mu = x_1$, $\Sigma = \sigma^2 I$, meaning a Gaussian centered on the conditioning data point x_1 with a minimal variance.

Conditional Flow Matching (CFM) (2)

- We define $u_t(x|x_1)$ as the velocity model that generates $p_t(x|x_1)$.
- From that the Conditional Flow Matching Loss is defined :

$$\mathcal{L}_{CFM}(\theta) = \mathbb{E}_{t, x_1 \sim q(x_1), x \sim p_t(x|x_1)} \left\| u_t^\theta(x) - u_t(x|x_1) \right\|^2 \quad (21)$$

- The theorem that enables the algorithm to be convenient states that minimizing $\mathcal{L}_{CFM}(\theta)$ is equivalent to minimizing $\mathcal{L}_{FM}(\theta)$:

$$\hat{\theta} = \min_{\theta} \mathcal{L}_{CFM}(\theta) = \min_{\theta} \mathcal{L}_{FM}(\theta) \quad (22)$$

- This objective is much more tractable, since we can derive the conditional velocity field $u_t(x|x_1)$ much easier than the marginal velocity field $u_t(x)$.

Solving the CFM optimization

- Now we focus on solving the minimization defined at Eq.(22).
- We put further constraints on $p_t(x|x_1)$; we now want it to be a Gaussian Probability Path. That means that, for every $t \in [0, 1]$:

$$p_t(x|x_1) = \mathcal{N}(\mu_t(x_1), \sigma_t(x_1)^2 I) \quad (23)$$

- $\mu_t(x_1)$ and $\sigma_t(x_1)$ are time dependent functions that have the role of transitioning from $p_0(x|x_1) = \mathcal{N}(0, 1)$ to $p_1(x|x_1) = \mathcal{N}(x_1, \sigma^2 I)$. (See Eq. (19), (20)).
- Note that path defined at Eq. (23) can be obtained by the flow (by linear transformation of a normal RV):

$$\phi_t(x_0) = \sigma_t(x_1)x_0 + \mu_t(x_1) \quad (24)$$

where x_0 is our initial gaussian noise and x_1 a data.

Solving the CFM optimization (2)

- Since $X_t = \phi_t(X_0) \sim p_t(x|x_1)$ for any $t \in [0, 1]$ if we assume $X_0 \sim \mathcal{N}(0, 1)$, then, from Eq. (3) we can derive the conditional velocity field $u_t(x|x_1)$ that generates $p_t(x|x_1)$ as :

$$u_t(x|x_1) = u_t(\phi_t(x_0)|x_1) = \frac{d}{dt}\phi_t(x_0) \quad (25)$$

- This way the CFM objective becomes :

$$\mathcal{L}_{CFM}(\theta) = \mathbb{E}_{t, x_1 \sim q(x_1), x \sim p(x_0)} \left\| u_t^\theta(\phi_t(x_0)) - \frac{d}{dt}\phi_t(x_0) \right\|^2 \quad (26)$$

Optimal Transport

- To derive $\phi_t(x_0)$ we need to choose some appropriate $\mu_t(x_1)$ and $\sigma_t(x_1)$ functions .
- A reasonable choice is changing the parameters linearly; the functions will be:

$$\mu_t(x_1) = tx_1 \quad (27)$$

$$\sigma_t(x_1) = 1 - (1 - \sigma_{min})t \quad (28)$$

- Note that, if we plug those into Eq. (23) we will satisfy the boundary conditions for $t = 0$ and $t = 1$ (Eq. (19) and (20)).
- The flow $\phi_t(x_0)$ at Eq. (24) defined that way is called Optimal Transport map:

$$\phi_t(x_0) = (1 - (1 - \sigma_{min})t)x_0 + tx_1 \quad (29)$$

Optimal Transport Loss

- We can easily get the derivative of $\phi_t(x_0)$:

$$\frac{d}{dt}\phi_t(x_0) = x_1 - (1 - \sigma_{min})x_0 \quad (30)$$

- Hence, we get the Optimal Transport Conditional Flow Matching Loss as :

$$\mathcal{L}_{CFM}^{OT}(\theta) = \mathbb{E}_{t, x_1 \sim q(x_1), x_0 \sim p(x_0)} \left\| u_t^\theta(\phi_t(x_0)) - (x_1 - (1 - \sigma_{min})x_0) \right\|^2 \quad (31)$$

- This loss can be easily computed by choosing a small enough σ_{min} and sampling $t \sim \mathcal{U}[0, 1]$, $x_0 \sim \mathcal{N}(0, 1)$ and x_1 from the available dataset.